Chemical Theory II
Statistical Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Calculating Q_N
  - Simplification: non-interacting particles.
  - The molecular partition function: $q = \Sigma_i \exp(-\varepsilon_i / kT)$
  - Distinguishable particles: $Q_N = q^N$
  - Indistinguishable particles: $Q_N = q^N / N!$ (provided $q >> N$)
- Thermodynamic functions in terms of q
  - U is not affected by (in)distinguishability.
  - S is greater for distinguishable particles.
- Calculating q
  - Separate molecular energy: translation + rotation + vibration + electronic
  - q factorises into a product of contributions.
  - Shift in energy origin: $q = q' \exp(-\varepsilon^0/kT)$

Evaluating *q* – the importance of *kT*

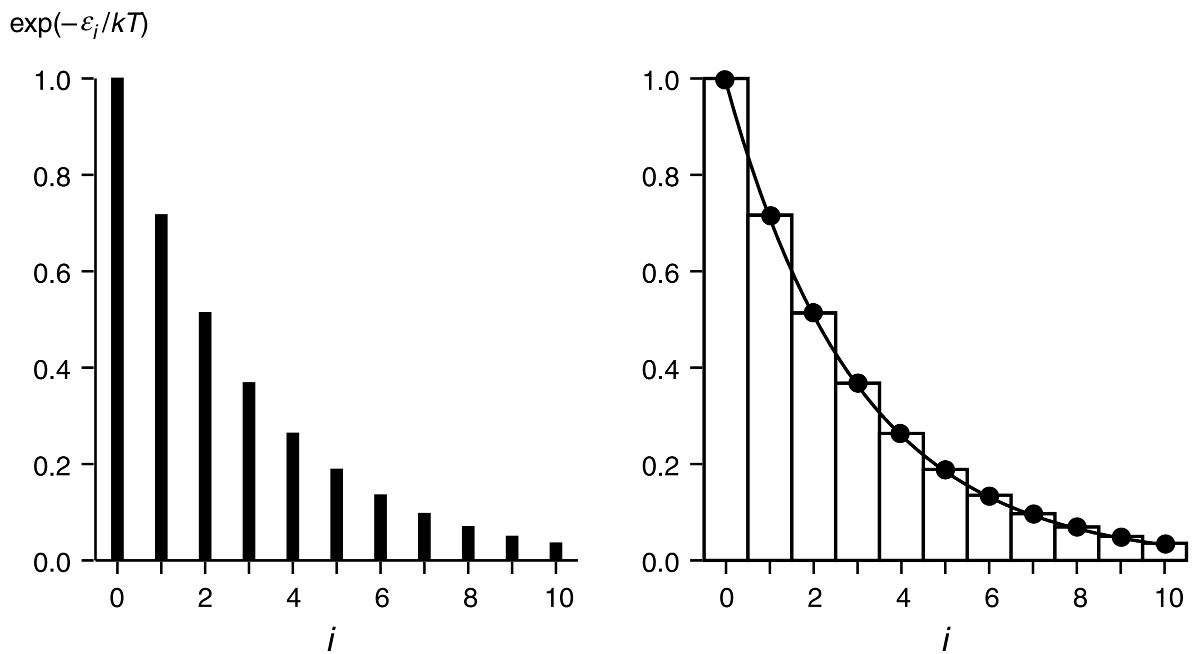
- As before, it is clear that once ***iΔ*** becomes significantly greater than kT, the term in the sum will not be significant. There are two limiting cases:
  - *Spacing much greater than **kT**: (**Δ >> kT**) in this case the second term is already insignificant and even more so the third and subsequent terms. The partition function is = 1*.
  - *Spacing much less than **kT**: (**Δ << kT**) in this case many terms will contribute to the sum (up to the term with **iΔ ≈ kT**). The partition function will be significantly greater than 1*.
- In the case that the spacing is comparable to ***kT***, a few terms will contribute and the partition function will be a bit over 1. In summary:
  - *If the energy level spacing significantly exceeds **kT**, only the ground state term contributes. If the spacing is much less than **kT**, many terms contribute*.
- From the Boltzmann distribution we see that energy levels which contribute to ***q***, i.e. those with $\boldsymbol{\varepsilon_i \leq kT}$, are those levels which will be populated. We can therefore identify ***q** as the number of accessible states*.

Evaluating *q* by using an integral

- Let us consider again the example of a set of energy levels which are evenly spaced; the partition function is

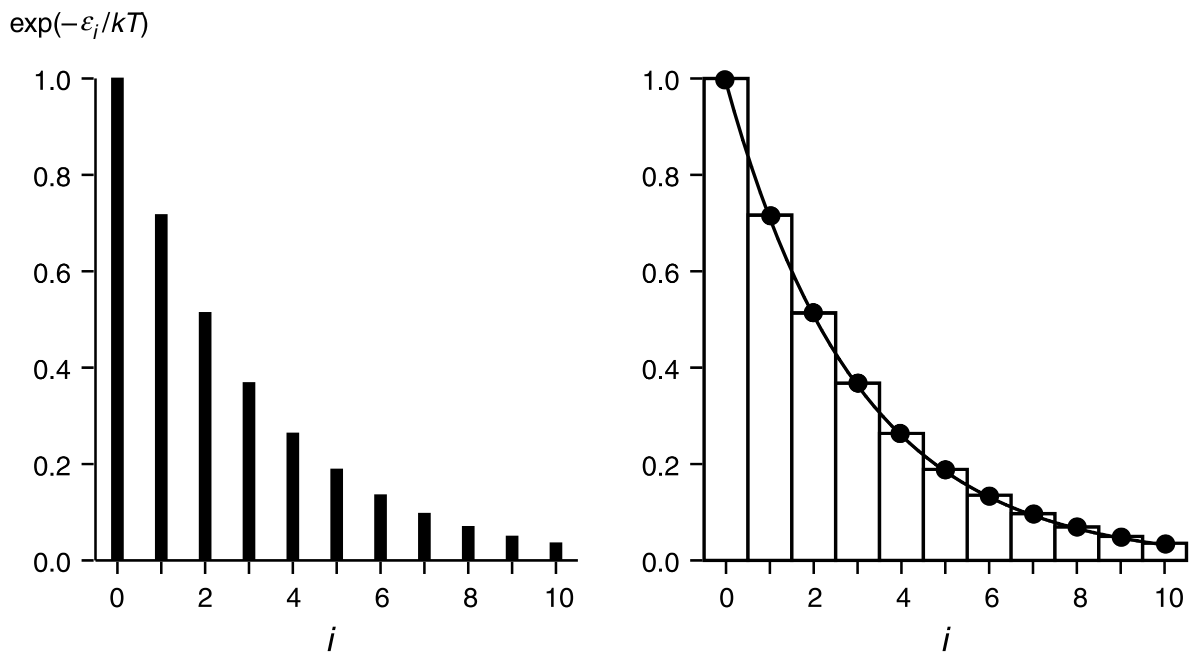
$$\begin{aligned} q' &= 1 + \exp\left(\frac{-\Delta}{kT}\right) + \exp\left(\frac{-2\Delta}{kT}\right) + \exp\left(\frac{-3\Delta}{kT}\right) + \ldots \\ &= \sum_{i=0}^{\infty} \exp\left(\frac{-i\Delta}{kT}\right). \end{aligned}$$

- Diagrammatically, each term can be represented as a bar of height ***exp (–iΔ / kT)***; the sum is obtained by adding the heights of the bars; this is illustrated in the plot below and on the left.



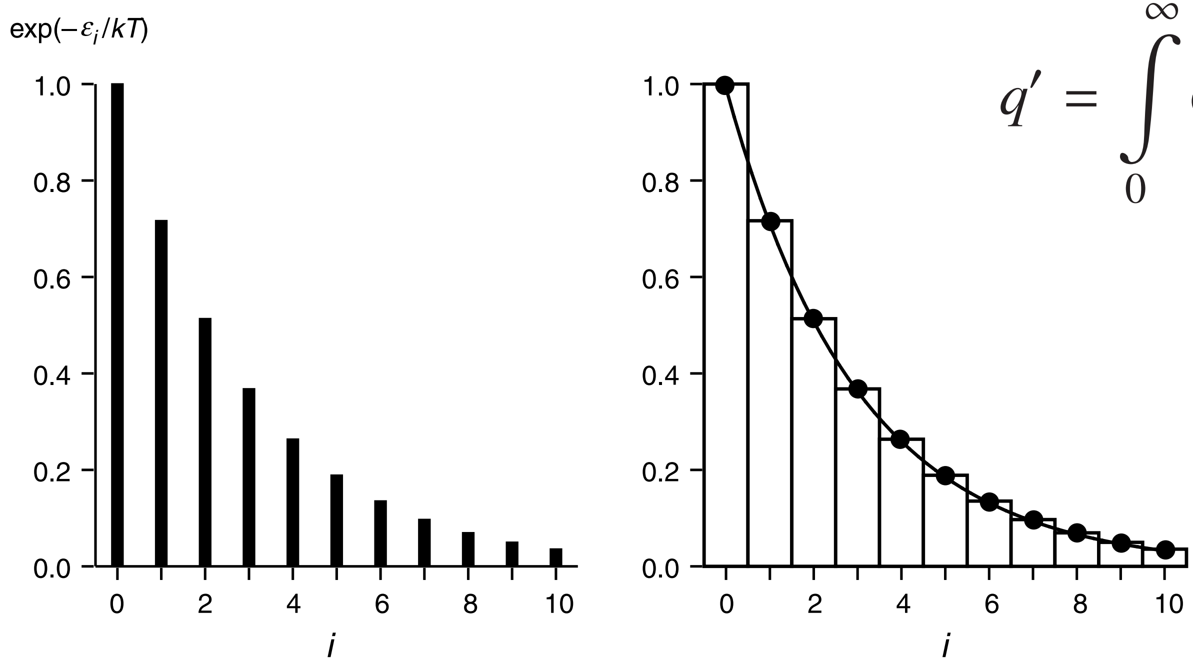
Evaluating *q* by using an integral

- A value of ***Δ*** has been chosen so that there are a significant number of terms contributing to the sum. In this situation the sum can be well approximated by an integral, as shown on the right. Here, the solid line is the function $\boldsymbol{\exp(-i\Delta / kT)}$ where $\boldsymbol{i}$ has been treated as a continuous variable, rather than an integer. The values of this function corresponding to the heights of the bars on the left are shown by dots. The open rectangles on the right have unit width and height equal to the bars on the left.



Evaluating *q* by using an integral

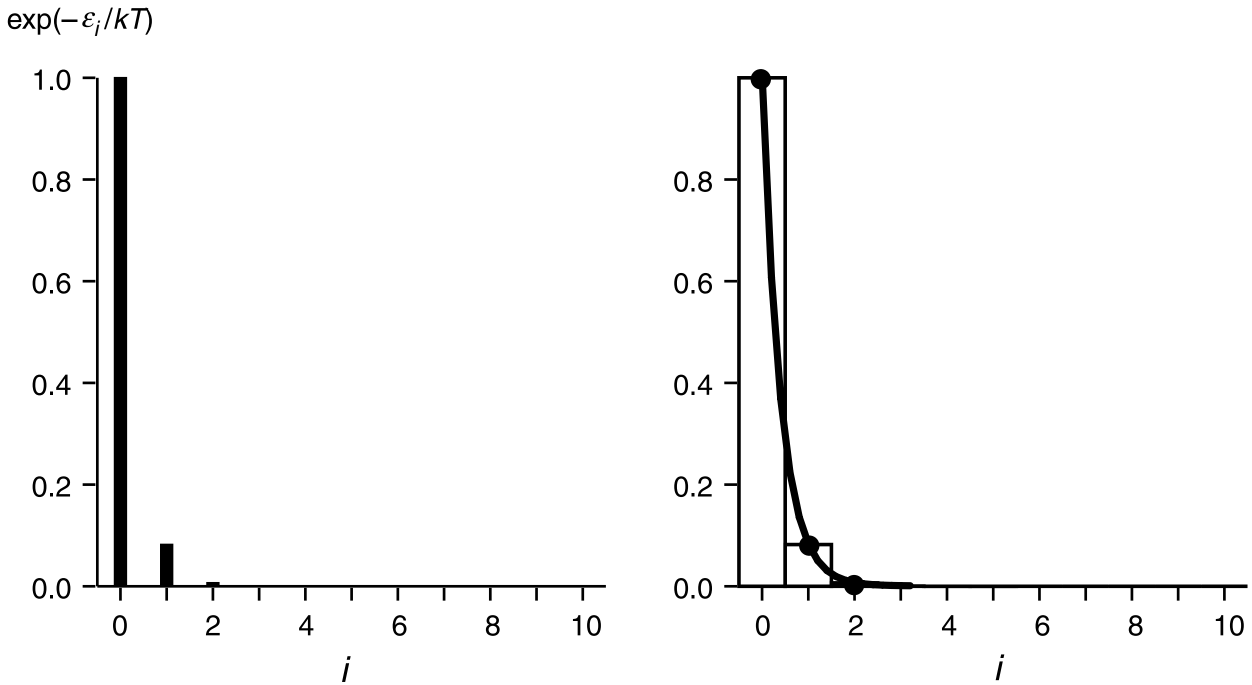
- It is clear from the plot that the sum of the areas of the open rectangles is equal to the required partition function. The area under the smooth curve is a reasonable approximation to the required sum. It is not exact, as there are little ‘triangles’ either included or excluded in the integral. The more and more rectangles which contribute, the closer and closer the integral comes to the sum.
- In summary, if there are many terms contributing to the partition function, the sum can be approximated by an integral. In this case



$$q' = \int_0^\infty \exp\left(\frac{-i\Delta}{kT}\right) \mathrm{d}i.$$

Evaluating *q* by using an integral

- On the other hand, if the spacing Δ is such that few terms contribute to the sum, the integral is a very poor approximation to the sum. This is illustrated below
- Now, it is clear that the area under the curve on the right is a very poor approximation to the sum of the areas of the rectangles. The integral cannot be used to calculate q'.



Example

- Consider a set of evenly spaced energy levels with spacing Δ i.e. 0, Δ, 2Δ, 3Δ . . . . Evaluate the partition function for the case that $\Delta = 0.05$ kT and then with $\Delta =$ kT .

Evaluating *q* by using an integral

- Recalling the discussion above, we noted that the situation in which many terms contribute to the partition function is when the energy level spacing is less than ***kT***. In summary
  - *If the energy level spacing is much less than* **kT**, *many terms contribute to the partition function and its value will be well approximated by an appropriate integral. Under these circumstances, the partition function must necessarily be much greater than 1*.

Evaluating *q* by using an integral

- We will find that it is quite often possible to evaluate the partition function using an integral. The result is a simple closed expression which is easier to handle than a sum; also, it will be possible to calculate ***dq / dT*** relatively simply (the derivative is needed to compute thermodynamic functions).
- If the energy level spacing is large compared to ***kT***, then we usually have no choice but to calculate ***q*** term by term. This is not as difficult as it sounds as in these circumstances only a few terms will be significant. The derivative with respect to T can also be calculated term by term:

$$\begin{aligned}\left(\frac{\partial q}{\partial T}\right)_V &= \frac{\partial}{\partial T}\left(\sum_i \exp\left(\frac{-\varepsilon_i}{kT}\right)\right) \\ &= \frac{1}{kT^2}\sum_i \varepsilon_i \exp\left(\frac{-\varepsilon_i}{kT}\right).\end{aligned}$$

Example

- Consider a set of evenly spaced energy levels with spacing Δ i.e. $0, \Delta, 2\Delta, 3\Delta \ldots$. Evaluate the partition function for the case that $\Delta = 0.05\ kT$ and then with $\Delta = kT$.
- For the two cases described above, calculate $(\partial q / \partial T)_V$

Degeneracy

- Some systems show *degeneracy, which is when there are distinct wavefunctions which have the same energy*. A simple example of this is the particle in a two-dimensional square box with side of length $\boldsymbol{a}$ whose energies are
$$E_{n_x,n_y} = \frac{h^2}{8ma^2}\left(n_x^2 + n_y^2\right).$$
- When $\boldsymbol{n_x}$ and $\boldsymbol{n_y}$ are different, for example $\boldsymbol{n_x = n_1}$, $\boldsymbol{n_y = n_2}$, there are always two degenerate wavefunctions, one with $\boldsymbol{n_x = n_1}$, $\boldsymbol{n_y = n_2}$ and one with $\boldsymbol{n_x = n_2}$, $\boldsymbol{n_y = n_1}$.
- In the definition of $\boldsymbol{q}$, the sum over $\boldsymbol{i}$ is a sum over states; each wavefunction counts as a separate state, even if it has the same energy as another wavefunction i.e. if there is degeneracy. For systems that show degeneracy, it is more convenient to define $\boldsymbol{q}$ as the sum over levels, $\boldsymbol{j}$:
$$q = \sum_j g_j \exp\left(\frac{-\varepsilon_j}{kT}\right)$$
where $\boldsymbol{\varepsilon_j}$ is the energy of the $\boldsymbol{j}$th level and *$\boldsymbol{g_j}$ is the degeneracy of that level i.e. the number of states of the same energy that are contained within that level*.

Course outline

- Partition functions in particular cases

Partition functions in particular cases

- In the following, partition functions for particular sets of energy levels will be evaluated. Often, we will be able to use a simple quantum mechanical model for the energy levels and so obtain simple results for the partition functions. In other cases, we will have to resort to finding the energy levels directly from spectroscopy.

Translational partition function

- The model used to find the translational energy levels is the particle in a box. Essentially what we are saying is that walls of a container which confine a gas are well approximated by an infinite potential barrier. The gas molecules cannot get out of the container, just as the particle cannot leave the box.
- In one dimension the energy levels, $\boldsymbol{E_n}$, are

$$E_n = \frac{n^2 h^2}{8ma^2}$$

where ***a*** *is the length of the box,* ***m*** *is the mass of the particle, and* ***n*** *is a quantum number which takes values 1, 2, 3 . . .* The first thing to decide is whether there are a large number of levels which will contribute in the evaluation of $\boldsymbol{q_{trans}}$; if so, the sum can be replaced by an integral.

Translational partition function

- One approach is to find the value of ***n***, n_{max}, for the energy level whose energy is ***kT***. According to the discussion earlier this give us an estimate of how many terms contribute:

$$\frac{n_{\max}^2 h^2}{8ma^2} = kT \qquad \text{hence} \qquad n_{\max}^2 = \frac{8ma^2 kT}{h^2}.$$

- For He atoms at room temperature and in a box of length 10 cm, n_{max} turns out to be about $\mathbf{2 \times 10^9}$. There are therefore an enormous number of energy levels which contribute to the sum in q_{trans} (i.e. *the translational energy levels are very closely spaced*), so it is *entirely safe to replace the sum by an integral*

$$q_{\text{trans,1D}} = \sum_{n=1}^{\infty} \exp\left(\frac{-E_n}{kT}\right) = \int_0^{\infty} \exp\left(\frac{-E_n}{kT}\right) \mathrm{d}n.$$

Translational partition function

- The lower limit of ***n*** in the sum is 1, but in the integral this limit has been replaced by 0. The reason for this is that it will be much easier to evaluate the integral with this limit and, in any case, with so many energy levels contributing, adding one extra will be of absolutely no consequence. The hypothetical level with n = 0 has zero energy, so no correction for the energy of the ground state is therefore needed.
- The integral is
$$\int_0^\infty \exp\left(\frac{-E_n}{kT}\right) dn = \int_0^\infty \exp\left(\frac{-n^2h^2}{8ma^2kT}\right) \mathrm{d}n$$
which is related to a standard form
$$\int_{-\infty}^{\infty} \exp\left(-\alpha x^2\right) \mathrm{d}x = \sqrt{\pi/\alpha} \quad \text{here} \quad \alpha = \frac{h^2}{8ma^2kT}$$

Translational partition function

- The integral we require is in the interval 0 to ∞ not –∞ to ∞; however, as the function is symmetrical about x = 0 the integral between 0 and ∞ is just half that between –∞ to ∞. So

$$\boxed{q_{\text{trans,1D}} = \frac{1}{2}\left(\frac{8mkTa^2\pi}{h^2}\right)^{\frac{1}{2}} = \left(\frac{2\pi mkT}{h^2}\right)^{\frac{1}{2}} a.}$$

- For the case of He in a 10 cm box, this translational partition function is about ***2×10⁹***; the large value is expected on account of the very large number of levels with energies less than ***kT***.

Translational partition function

$$\boxed{q_{\text{trans,1D}} = \frac{1}{2}\left(\frac{8mkTa^2\pi}{h^2}\right)^{\frac{1}{2}} = \left(\frac{2\pi mkT}{h^2}\right)^{\frac{1}{2}} a.}$$

- The expression for ***q***

- The expression for $\boldsymbol{q_{trans}}$ in the equation above is often written in terms of *the thermal wavelength,* ***Λ***, defined as

$$\Lambda = \left(\frac{h^2}{2\pi mkT}\right)^{\frac{1}{2}}$$

giving

$$q_{\text{trans,1D}} = \frac{a}{\Lambda}.$$

As its name implies, ***Λ*** *has dimensions of length.*

- For the case of He in the 10 cm box, ***Λ*** = *50.6 pm*.

Translational partition function

- For a *three dimensional* box, the energy is just the sum of contributions from the x, y and z dimensions. Three quantum numbers are needed, giving the energy as

$$E_{n_x,n_y,n_z} = \frac{h^2}{8ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)$$

- where it has been assumed that the box is of side ***a*** in all three dimensions. As the energy levels are additive, the partition functions are multiplicative, so

$$\begin{aligned} q_{\mathrm{trans,3D}} &= q_{\mathrm{trans,1D},x}\; q_{\mathrm{trans,1D},y}\; q_{\mathrm{trans,1D},z} \\ &= \left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} a^3 \\ &= \left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V, \end{aligned}$$

- where $\boldsymbol{a^3}$ has been replaced by the volume, ***V***. In terms of the thermal wavelength, $\boldsymbol{q_{trans,3D}}$ *is simply* $(\boldsymbol{V} / \boldsymbol{\Lambda}^3)$.
- Typically $\boldsymbol{q_{trans,3D}}$ *is of the order of* 10^{30}

Translational partition function

- As the translational energy levels change with the dimensions of the box (the volume), so does the value of the partition function. *It is for this reason that it is convenient for us to work at fixed volume (rather than fixed pressure) as this keeps the energy levels fixed*. So, in our discussion of the approach to chemical equilibrium we shall *use the Helmholtz energy, A, rather than G*.

Example

- Calculate the thermal wavelength, Λ, and the partition function for translation in three dimensions for a sample of N2 gas at 298K held in a volume of 1 dm3.

Rotational partition function for diatomics

- The model we use for the rotational levels of a diatomic molecule is the rigid rotor. This has energies $\boldsymbol{E_J}$
  $\boldsymbol{E_J = B\,J\,(J+1)}$
- where *J is the rotational quantum number, taking values 0, 1, 2 . . .* ; note that each level has a degeneracy of ***(2J + 1)***. *B is the rotational constant*, given by

$$B = \frac{\hbar^2}{2I}$$

- where *I is the moment of inertia*. For a diatomic molecule $\boldsymbol{I = \mu R^2}$; *R is the bond length and μ is the reduced mass* given by

$$\mu = \frac{m_1 m_2}{(m_1 + m_2)}$$

- where *m_1 and m_2 are the masses of the two atoms*. If the energies and rotational constant are expressed in wavenumbers, cm^{-1}, then

$$\tilde{E}_J = \tilde{B}J\,(J+1) \qquad \tilde{B} = \frac{h}{8\pi^2\tilde{c}I} \qquad \tilde{c} \text{ in cm s}^{-1}$$

Rotational partition function for diatomics

- The rotational partition function is,

$$q_{\text{rot}} = \sum_{J=0}^{\infty} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right)$$

and the first few terms are

$$q_{\text{rot}} = 1 + 3\exp\left(\frac{-2B}{kT}\right) + 5\exp\left(\frac{-6B}{kT}\right) + \dots .$$

- The energy of the ground state (*J = 0*) is zero, so no correction for the energy zero is needed. As before, we would like to know if it is permissible to evaluate the sum as an integral. To do this, we need to compare the energy level spacings with ***kT***; if they are smaller that ***kT***, we may safely use an integral.
- The energies of the rotational levels are in the series *0, 2B, 6B . . .*, so we can say that the spacings of the levels are of the order of B. For diatomic molecules, B varies from a maximum of about 60 cm^{-1} for H_2, about 2 cm^{-1} for CO to less than 0.1 cm^{-1} for heavy diatomics such as I_2.

Rotational partition function for diatomics

- To compare ***B*** to ***kT*** we must have both in the same units, say *Joules*. However, as rotational constants (like many other spectroscopic parameters) are often quoted in cm^{-1} it would be more convenient to know what ***kT*** is in cm^{-1}. All that matters in our evaluation of ***q*** is the dimensionless ratio ***B / kT***; so long as both ***B*** and ***kT*** are in the same units, the actual units are irrelevant.
- The conversion from Joules to cm^{-1} is

$$\frac{\text{Joules}}{h \times \tilde{c}} = \text{cm}^{-1} \qquad \tilde{c} \text{ in cm s}^{-1}$$

- In SI k is in J K^{-1}, so converting to cm^{-1} K^{-1} simply means dividing by **h** $\tilde{\mathbf{c}}$. The result is k = 0.6950 cm^{-1} K^{-1}. Thus *at room temperature (298 K),* ***kT*** *is 0.695 ×298 = 207* cm^{-1}.

Rotational partition function for diatomics

- So, even for H_2 which has the largest rotational constant, ***kT*** *is significantly in excess of* ***B*** *at room temperature. Several terms will therefore contribute to the partition function, and so it is quite justified to use an integral to approximate the sum. For diatomics with the largest rotational constants, this approximation will only begin to break down at temperatures significantly below room temperature*.
- Assuming that an integral can be used, the rotational partition function is given by

$$q_{\mathrm{rot}} = \int_0^\infty (2J+1)\exp\left(\frac{-BJ(J+1)}{kT}\right)\mathrm{d}J.$$

- The integral is found by noting that

$$\frac{\mathrm{d}}{\mathrm{d}J}\exp\left(\frac{-BJ(J+1)}{kT}\right) = \frac{-B(2J+1)}{kT}\exp\left(\frac{-BJ(J+1)}{kT}\right)$$

Rotational partition function for diatomics

- So that
$$q_{\mathrm{rot}} = \left[\frac{-kT}{B}\exp\left(\frac{-BJ(J+1)}{kT}\right)\right]_0^\infty = \frac{kT}{B}.$$
- It is common to define *a rotational temperature, **θ_{rot}**,* as $\boxed{\theta_{\mathrm{rot}} = \frac{B}{k};}$

  the name derives from the fact that ***θ_{rot}*** *has the dimension of temperature*. The rotational partition function can thus be written:

$$\boxed{q_{\mathrm{rot}} = \frac{kT}{\sigma B} \quad \text{or} \quad q_{\mathrm{rot}} = \frac{T}{\sigma\theta_{\mathrm{rot}}}.}$$

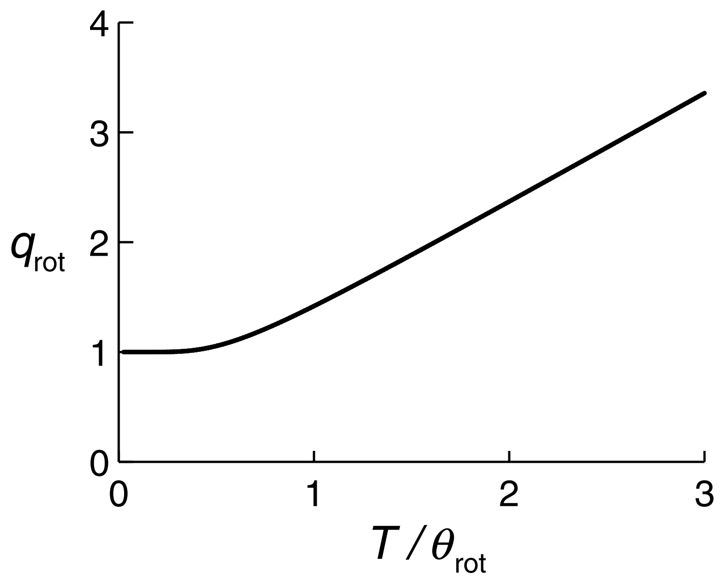
  Where the *symmetry parameter, **σ**,* has been introduced. *It takes the value 2 for homonuclear diatomics and 1 for heteronuclear diatomics*.
- For CO, with $\tilde{B}$ = 1.925 cm^{-1}, the rotational temperature is easily calculated if we use the value of k in cm^{-1} K^{-1} (k = 0.6950 cm^{-1} K^{-1}). Hence ***θ_{rot}*** = 1.925/0.695 = 2.77 K and ***q_{rot}*** = 108 at 298 K. *The rotational partition function is very much smaller than the translational partition function; nevertheless the fact that many rotational states contribute means that **q_{rot}** is significantly greater than 1*.

Example

- The rotational constant of I_2 is 0.0373 cm–1; compute the rotational partition function at 298 K.

Rotational partition function for diatomics

- *If it is not the case that* $\boldsymbol{T > \theta_{rot}}$*, the equation derived from integration cannot be used and the rotational partition function has to be evaluated by computing successive terms in the sum until they are insignificant. In practice only a few terms are needed as in this limit only a few levels are occupied. Shown below is a plot of* $\boldsymbol{q_{rot}}$ *as a function of* $\boldsymbol{T / \theta_{rot}}$*; as expected the partition function is 1 when* $\boldsymbol{T / \theta_{rot} \ll 1}$*. The value of* $\boldsymbol{q_{rot}}$ *rises quickly once* $\boldsymbol{T / \theta_{rot} \approx 1}$ *and at higher temperatures the partition function rises linearly with* T *as predicted by the formula from integration*.
- The rotational energy levels of linear polyatomic molecules are identical to those for a diatomic; however the expression for the moment of inertia is more complex.



Rotational partition function for non-linear molecules

- Non-linear molecules have three moment of inertia, denoted ***I_a***, ***I_b*** and ***I_c***. Associated with each is a rotational constant, for example

$$\tilde{A} = \frac{h}{8\pi^2 \tilde{c} I_a} \qquad \tilde{c} \text{ in cm s}^{-1}, \quad I_a \text{ in kg m}^2$$

- In turn each of the rotational constants is associated with a rotational temperature, for example

$$\theta_{\mathrm{rot},a} = \frac{\tilde{A}}{k}.$$

- In the high temperature limit where ***$T >> \theta_{rot}$*** it can be shown that the rotational partition function is given by

$$q_{\mathrm{rot}} = \frac{\pi^{\frac{1}{2}}}{\sigma} \left(\frac{T^3}{\theta_{\mathrm{rot},a}\, \theta_{\mathrm{rot},b}\, \theta_{\mathrm{rot},c}} \right)^{\frac{1}{2}}$$

where ***σ*** is the symmetry number. *For H_2O σ is two, and for NH_3 **σ** is three*.